TECHNICAL DATA AUDIT & EPIDEMIOLOGICAL SCREENING REPORT

Waterborne Toxicants & Fetal Mortality in Yucatán (2012–2025)

Integration, Data Quality Remediation, and Municipal Rate Normalization across CONAGUA & Vital Statistics Datasets

1,323 → 1,179
Water samples filtered to 29 high-risk Yucatán municipalities (2012–2025)

32 / 208
Water parameters linked to infant mortality & low birth weight

7.34 vs 1.32
2012 mean fetal deaths/mun: flagged (29) vs. unflagged (77)

12 / 29
Flagged municipalities exceeding 2025 state death-to-birth mean (1.77%)

1. Datasets Utilized

Three primary datasets were ingested, audited, and cross-referenced in Python (`pandas`) to evaluate spatial overlap between toxic water contaminants and adverse perinatal outcomes in Yucatán, Mexico:

- **CONAGUA Water Quality Monitoring (`RESULTADOS-YUCATÁN.xlsx`):** Multi-sheet workbook covering 2012–2025 across Yucatán. Contains `SITIOS` (151 monitoring stations with coordinates, aquifer, and municipality), `RESULTADOS` (1,323 sampling events across 208 physicochemical and microbiological parameters), and `SIMBOLOGÍA` (parameter definitions and units).
- **National Fetal Deaths Microdata 2012 (`fetales_2012_unificado.csv`):** 22,758 national fetal death records (95 variables) detailing maternal residence (`ent_resid`, `mun_resid`), occurrence location, gestational age, birth weight, and CIE-10 cause of death.
- **Yucatán Municipal Births & Fetal Deaths 2025 (`nacimientos_y_defunciones_fetales_yucatan_2025.csv`):** Complete municipal-level summary covering all 106 municipalities of Yucatán (19,897 live births and 329 fetal deaths).

2. Data Preparation & Quality Remediation

Raw environmental and vital statistics files exhibited structural inconsistencies that required systematic preprocessing prior to statistical comparison:

Data Quality Issue	Diagnostic Finding	Remediation Applied
Censored Detection Limits	132 of 208 water columns loaded as text (`object`) due to < or > prefixes (e.g., `<25`, `<0.005`).	Parsed via regex: substituted below-limit values $<X$ with $0.5 \times X$ (standard environmental half-limit) and $>X$ with X as `float64`.
Municipality Name Splits	Inconsistent accents and trailing punctuation split identical municipalities (e.g., `MERIDA` vs. `MÉRIDA`, `UMAN` vs. `UMÁN`, `TIXKOKOB.`).	Applied Unicode NFD normalization, stripped diacritics/ trailing dots, and standardized to uppercase across all datasets.
High Matrix Sparsity	87 water parameters had >90% nulls because coastal/wetland vs. groundwater stations test different panels.	Retained all 32 perinatal-relevant toxicants; screened municipalities using 9 core parameters with 44%–93% completeness.
Confidential Locality Codes	In 2012 microdata, 99 of 317 Yucatán records masked locality names under SNIEG confidentiality rules.	Mapped numeric INEGI codes (`ent_resid = 31`, `mun_resid` 1–106) against the official 106-municipality catalog, recovering 99.4% of residences (only 2 coded 999).

3. Analytical Procedures

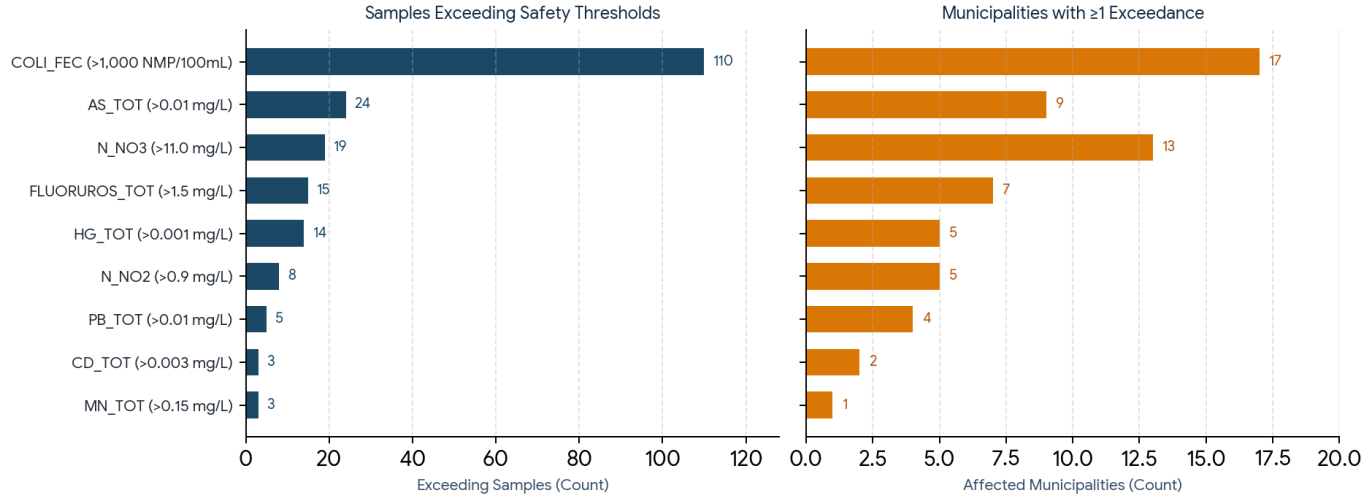
1. **Toxicant Selection & Threshold Screening:** Selected 32 water contaminants linked in epidemiological literature to fetal growth restriction, low birth weight, or infant mortality: nitrogen compounds (`N_NO3`, `N_NO2`), heavy metals/metalloids (`AS_TOT`, `PB_TOT`, `HG_TOT`, `CD_TOT`, `MN_TOT`), fecal pathogens (`COLI_FEC`, `ENTEROC_FEC`, `E_COLI`), disinfection byproducts (`THMs`, haloacetic acids), VOCs (`BENCENO`, `TRICLOROETILENO`), fluorides, and cyanides. Flagged any municipality exceeding Mexican potable water limits (NOM-127-SSA1-2021) or $COLI_FEC > 1{,}000$ NMP/100 mL.
2. **2012 Absolute & Share Benchmarks:** Filtered national microdata to maternal residence in Yucatán (317 cases; 1.39% of 22,744 national cases across 32 states) and evaluated flagged municipalities against four benchmarks: state municipal mean across all 106 municipalities (2.97 deaths; 0.94%), state mean across 60 active municipalities (5.25 deaths; 1.67%), national mean across 2,457 municipalities (9.25 deaths; 1.30%), and national mean across 1,666 active municipalities (13.63 deaths; 1.92%).

3. **2025 Population-Normalized Rate Computation:** To eliminate urban population bias, computed the municipal fetal death-to-birth ratio $R_i = (Fetal\ Deaths_i / Live\ Births_i) \times 100$ for 2025 and benchmarked each flagged municipality against the Yucatán 106-municipality mean (**1.77%**) and the 29-municipality flagged group mean (**1.74%**).

4. Results: Water Quality & Perinatal Mortality

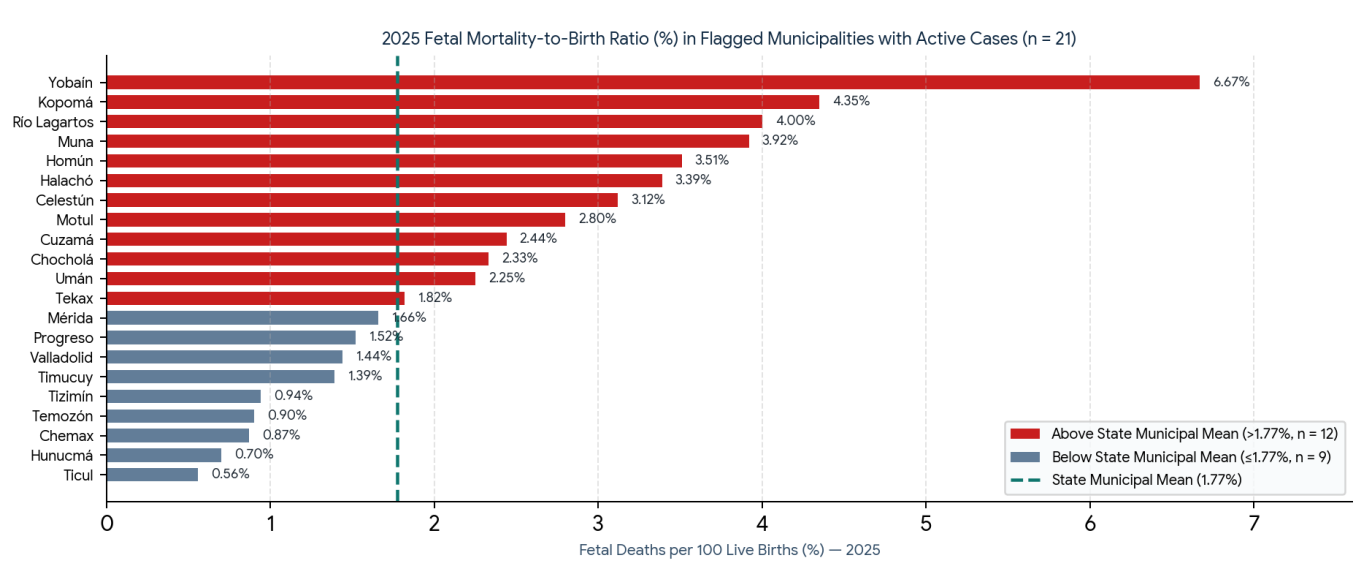
4.1 High-Risk Municipalities Identified in Water Monitoring (2012–2025)

Exactly **29 of 70 monitored municipalities** exceeded at least one health threshold, accounting for **1,179 of 1,323** total sampling records. Fecal coliforms (110 samples across 17 municipalities), total arsenic (24 samples across 9 municipalities, max 0.2064 mg/L in Mérida), and nitrates (19 samples across 13 municipalities, max 39.63 mg/L in Tizimín) drove the majority of exceedances.



4.2 Comparison of 2012 Counts vs. 2025 Birth-Normalized Ratios

In 2012, the 29 flagged municipalities accounted for **213 of 317** state fetal deaths (67.2%), averaging **7.34 deaths/municipality** compared to **1.32** in the remaining 77 municipalities. Four populous municipalities—**Mérida (109), Tizimín (19), Chemax (15), and Valladolid (15)**—exceeded every state and national count benchmark. However, normalizing by 2025 live births reveals a sharp structural shift: large urban centers fall below the 1.77% state municipal average, whereas smaller rural and coastal municipalities with nitrate, arsenic, or fecal contamination exhibit the highest relative fetal loss rates.



Municipality	Primary Exceeded Toxicants	2012 Deaths	2025 Births	2025 Deaths	2025 Ratio (%)	> State Mean (1.77%)
YOBAIN	COLI_FEC (max 24,196)	2	15	1	**6.67%**	Yes
KOPOMA	N_NO2, AS_TOT, COLI_FEC	0	23	1	**4.35%**	Yes

Municipality	Primary Exceeded Toxicants	2012 Deaths	2025 Births	2025 Deaths	2025 Ratio (%)	> State Mean (1.77%)
RIO LAGARTOS	N_NO3, AS_TOT, PB_TOT, HG_TOT, COLI_FEC	0	25	1	**4.00%**	Yes
MUNA	N_NO3 (max 17.40 mg/L)	2	102	4	**3.92%**	Yes
HOMUN	AS_TOT, COLI_FEC (E. coli 11,000)	1	57	2	**3.51%**	Yes
HALACHO	N_NO3 (mean 16.31), FLUORUROS_TOT (4.14)	2	236	8	**3.39%**	Yes
CELESTUN	HG_TOT (max 0.003), FLUORUROS, COLI_FEC	2	64	2	**3.12%**	Yes
MOTUL	N_NO3 (mean 12.87 mg/L)	6	322	9	**2.80%**	Yes
CUZAMA	AS_TOT (max 0.1139), COLI_FEC	0	41	1	**2.44%**	Yes
CHOCHOLA	N_NO3 (>11.0 mg/L)	1	43	1	**2.33%**	Yes
UMAN	N_NO2 (max 2.46), FLUORUROS_TOT	7	757	17	**2.25%**	Yes
TEKAX	N_NO3 (>11.0 mg/L)	7	440	8	**1.82%**	Yes
MERIDA	N_NO3, N_NO2, AS_TOT, HG_TOT, MN_TOT, FLUOR, COLI	109	7,149	119	1.66%	No
PROGRESO	N_NO2, PB_TOT, HG_TOT, CD_TOT, COLI_FEC	7	528	8	1.52%	No
VALLADOLID	N_NO3, AS_TOT (mean 0.0177), FLUORUROS_TOT	15	905	13	1.44%	No
TIZIMIN	N_NO3 (39.63), AS_TOT, PB_TOT, HG_TOT, COLI_FEC	19	848	8	0.94%	No

Note: Remaining 13 flagged municipalities below the 1.77% threshold in 2025: Timucuy (1.39%), Temozón (0.90%), Chemax (0.87%), Hunucmá (0.70%), Ticul (0.56%), and 8 municipalities with 0 fetal deaths (Chapab, Dzilam de Bravo, Espita, San Felipe, Telchac Puerto, Tepakán, Tixpéhual, Yaxkukul).

5. Key Findings & Methodological Limitations

Key Findings

- **Widespread Karst Vulnerability:** 41.4% of monitored municipalities (29/70) exceeded health limits at least once, led by fecal coliforms and agricultural/wastewater nitrates in the karst aquifer.
- **Population Distortion Corrected:** While Mérida concentrated 34.6% of Yucatán's 2012 fetal deaths (109/315), its 2025 death-to-birth ratio (1.66%) sits slightly below the state municipal mean (1.77%).
- **Persistent High-Risk Corridors: Umán, Tekax, Motul, and Halachó** stand out as priority targets—combining substantial birth cohorts (>230 births), elevated 2025 ratios (1.82%–3.39%), and chronic nitrate/nitrite or fluoride exceedances.

Methodological Limitations

- **Small-Denominator Volatility:** In municipalities with <30 annual births (e.g., Yobaín: 15 births; Kopomá: 23; Río Lagartos: 25), a single fetal death produces a 4.0%–6.7% ratio; multi-year pooling is required for statistical stability.
- **Ecological Fallacy & Exposure Proxy:** CONAGUA stations sample wells, cenotes, and coastal wetlands rather than household tap water; municipal aggregation cannot confirm individual maternal ingestion.
- **Temporal & Spatial Coverage Gaps:** 36 of Yucatán's 106 municipalities lacked CONAGUA stations, and water sampling frequency dropped sharply in 2021 (13 samples) and 2025 (7 samples).

Verification Package Included: Alongside this PDF, a reproducible verification archive (`verification_package_yucatan_2012_2025.zip`) and standalone CSVs are provided containing: (1) `01_water_contaminants_29_municipalities_2012_2025.csv` (1,179 cleaned water records), (2) `02_fetales_2012_yucatan_limpio.csv` (317 mapped 2012 records), (3) `03_fetales_2012_yucatan_municipios_toxicos.csv` (213 records in flagged municipalities), (4) `04_municipios_peligrosos_fetales_vs_nacimientos_2025.csv` (29-municipality 2025 rate audit), and (5) `reproduce_analysis.py` (end-to-end verification script).